

State-space models

Hidden Markov models

Dr. Jarad Niemi

Iowa State University

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Structure

Observation equation:

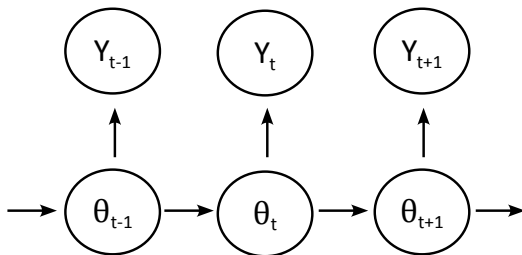
$$Y_t = f_t(\theta_t, v_t)$$

$$Y_t | \theta_t \stackrel{\text{ind}}{\sim} p_t(y_t | \theta_t, \dots)$$

State transition (evolution) equation:

$$\theta_t = g_t(\theta_{t-1}, w_t)$$

$$\theta_t | \theta_{t-1} \stackrel{\text{ind}}{\sim} p_t(\theta_t | \theta_{t-1}, \dots)$$



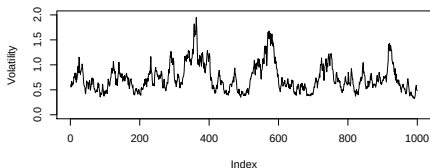
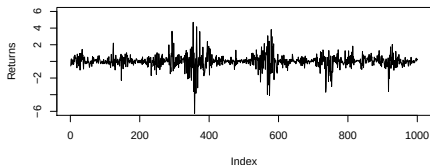
Notation and terminology

Observation equation:	$Y_t = f_t(\theta_t, v_t)$
Observations:	Y_t
Observation (measurement) error:	v_t
State transition (evolution) equation:	$\theta_t = g_t(\theta_{t-1}, w_t)$
Latent (unobserved) state:	θ_t
Evolution noise	w_t

Stochastic volatility model

$$y_t \sim N(0, \sigma_t^2)$$
$$\log \sigma_t \sim N(\mu + \phi[\log \sigma_{t-1} - \mu], W)$$

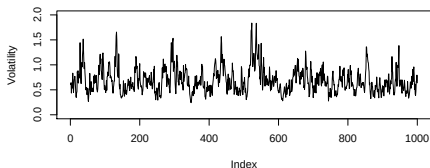
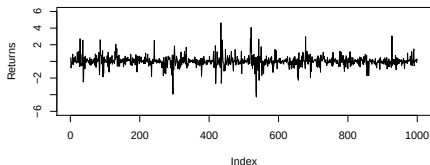
$\phi = 0.95, W = 0.1^2$



Stochastic volatility

$$y_t \sim N(0, \sigma_t^2)$$
$$\log \sigma_t \sim N(\mu + \phi(\log \sigma_{t-1} - \mu), W)$$

$\phi = 0.8, W = 0.2^2$

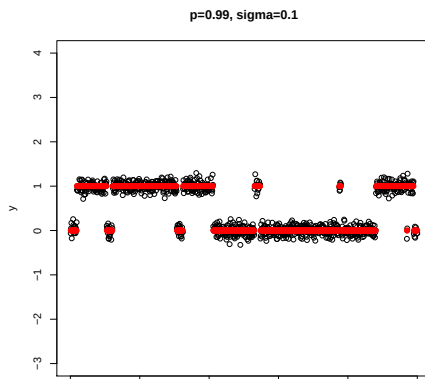


Markov switching model

$$y_t \sim N(\theta_t, \sigma^2)$$

$$\theta_t \sim p\delta_{\theta_{t-1}} + (1-p)\delta_{1-\theta_{t-1}}$$

$$\theta_0 = 0$$

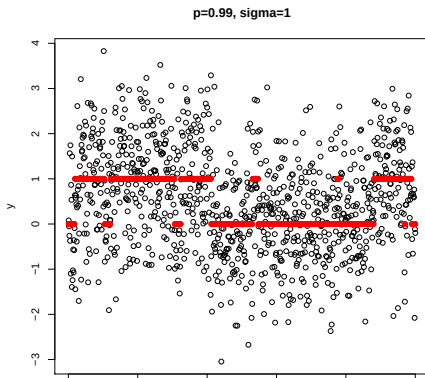


Markov switching model

$$y_t \sim N(\theta_t, \sigma^2)$$

$$\theta_t \sim p\delta_{\theta_{t-1}} + (1-p)\delta_{1-\theta_{t-1}}$$

$$\theta_0 = 0$$



Goals:

- Filtering
- Smoothing
- Forecasting

What do we know?

- $p(y_t|\theta_t)$ for all t
- $p(\theta_t|\theta_{t-1})$ for all t
- $p(\theta_0)$

In principle, we could have subscripts for the distributions/densities, e.g.

- $p_t(y_t|\theta_t)$ for all t
- $p_t(\theta_t|\theta_{t-1})$ for all t

to indicate that the form of the distribution/density has changed. But, in most models the form stays the same and only the state changes with time.

For simplicity, we will assume a time-homogeneous process and therefore remove the subscript.

Filtering

Goal: $p(\theta_t | y_{1:t})$ where $y_{1:t} = (y_1, y_2, \dots, y_t)$ (filtered distribution)

Recursive procedure:

- Assume $p(\theta_{t-1} | y_{1:t-1})$
- Prior for θ_t

$$\begin{aligned} p(\theta_t | y_{1:t-1}) &= \int p(\theta_t, \theta_{t-1} | y_{1:t-1}) d\theta_{t-1} \\ &= \int p(\theta_t | \theta_{t-1}, y_{1:t-1}) p(\theta_{t-1} | y_{1:t-1}) d\theta_{t-1} \\ &= \int p(\theta_t | \theta_{t-1}) p(\theta_{t-1} | y_{1:t-1}) d\theta_{t-1} \end{aligned}$$

- One-step ahead predictive distribution for y_t

$$\begin{aligned} p(y_t | y_{1:t-1}) &= \int p(y_t, \theta_t | y_{1:t-1}) d\theta_t \\ &= \int p(y_t | \theta_t, y_{1:t-1}) p(\theta_t | y_{1:t-1}) d\theta_t \\ &= \int p(y_t | \theta_t) p(\theta_t | y_{1:t-1}) d\theta_t \end{aligned}$$

- Filtered distribution for θ_t

$$p(\theta_t | y_{1:t}) = \frac{p(y_t | \theta_t, y_{1:t-1}) p(\theta_t | y_{1:t-1})}{p(y_t | y_{1:t-1})} = \frac{p(y_t | \theta_t) p(\theta_t | y_{1:t-1})}{p(y_t | y_{1:t-1})}$$

What do we know now?

- $p(y_t|\theta_t)$ for all t
- $p(\theta_t|\theta_{t-1})$ for all t
- $p(\theta_0)$
- $p(\theta_t|y_{1:t-1})$ for all t
- $p(y_t|y_{1:t-1})$ for all t

Smoothing

Goal: $p(\theta_t|y_{1:T})$ for $t < T$

- Backward transition probability $p(\theta_t|\theta_{t+1}, y_{1:t})$

$$\begin{aligned} p(\theta_t|\theta_{t+1}, y_{1:T}) &= p(\theta_t|\theta_{t+1}, y_{1:t}) \\ &= \frac{p(\theta_{t+1}|\theta_t, y_{1:t})p(\theta_t|y_{1:t})}{p(\theta_{t+1}|y_{1:t})} \\ &= \frac{p(\theta_{t+1}|\theta_t)p(\theta_t|y_{1:t})}{p(\theta_{t+1}|y_{1:t})} \end{aligned}$$

- Recursive smoothing distributions $p(\theta_t|y_{1:T})$ starting from $p(\theta_T|y_{1:T})$

$$\begin{aligned} p(\theta_t|y_{1:T}) &= \int p(\theta_t, \theta_{t+1}|y_{1:T})d\theta_{t+1} \\ &= \int p(\theta_{t+1}|y_{1:T})p(\theta_t|\theta_{t+1}, y_{1:T})d\theta_{t+1} \\ &= \int p(\theta_{t+1}|y_{1:T})\frac{p(\theta_{t+1}|\theta_t)p(\theta_t|y_{1:t})}{p(\theta_{t+1}|y_{1:t})}d\theta_{t+1} \\ &= p(\theta_t|y_{1:t}) \int \frac{p(\theta_{t+1}|\theta_t)}{p(\theta_{t+1}|y_{1:t})}p(\theta_{t+1}|y_{1:T})d\theta_{t+1} \end{aligned}$$

Forecasting

Goal: $p(y_{t+k}, \theta_{t+k} | y_{1:t})$

$$p(y_{t+k}, \theta_{t+k} | y_{1:t}) = p(y_{t+k} | \theta_{t+k}) p(\theta_{t+k} | y_{1:t})$$

Recursively, given $p(\theta_{t+(k-1)} | y_{1:t})$

$$\begin{aligned} p(\theta_{t+k} | y_{1:t}) &= \int p(\theta_{t+k}, \theta_{t+(k-1)} | y_{1:t}) d\theta_{t+(k-1)} \\ &= \int p(\theta_{t+k} | \theta_{t+(k-1)}, y_{1:t}) p(\theta_{t+(k-1)} | y_{1:t}) d\theta_{t+(k-1)} \\ &= \int p(\theta_{t+k} | \theta_{t+(k-1)}) p(\theta_{t+(k-1)} | y_{1:t}) d\theta_{t+(k-1)} \\ p(y_{t+k} | y_{1:t}) &= \int p(y_{t+k} | \theta_{t+k}) p(\theta_{t+k} | y_{1:t}) p\theta_{t+k} \end{aligned}$$

Filtering in a Markov switching model

$$\begin{aligned} y_t &\sim N(\theta_t, \sigma^2) \\ \theta_t &\sim p\delta_{\theta_{t-1}} + (1-p)\delta_{1-\theta_{t-1}} \\ \theta_0 &= 0 \end{aligned}$$

- Note: $p(\theta_t = 1) = 1 - p(\theta_t = 0)$ for all t
- Suppose $q = p(\theta_{t-1} = 1|y_{1:t-1})$. What is $p(\theta_t = 1|y_{1:t-1})$?

$$p(\theta_t = 1|y_{1:t-1}) = \sum_{k=0}^1 p(\theta_t = 1|\theta_{t-1} = k)p(\theta_{t-1} = k|y_{1:t-1}) = (1-p)(1-q) + pq = p_1$$

- What is $p(\theta_t = 0|y_{1:t-1})$?

$$p(\theta_t = 0|y_{1:t-1}) = \sum_{k=0}^1 p(\theta_t = 0|\theta_{t-1} = k)p(\theta_{t-1} = k|y_{1:t-1}) = p(1-q) + (1-p)q = p_0$$

- What is $p(y_t|y_{1:t-1})$?

$$p(y_t|y_{1:t-1}) = \sum_{k=0}^1 p(y_t|\theta_t = k)p(\theta_t = k|y_{1:t-1}) = p_0 N(y_t; 0, \sigma^2) + p_1 N(y_t; 1, \sigma^2)$$

- What is $p(\theta_t = 1|y_{1:t})$?

$$p(\theta_t = 1|y_{1:t}) = \frac{p(y_t|\theta_t = 1)p(\theta_t = 1|y_{1:t-1})}{p(y_t|y_{1:t-1})} = \frac{p_1 N(y_t; 1, \sigma^2)}{p_0 N(y_t; 0, \sigma^2) + p_1 N(y_t; 1, \sigma^2)}$$

Hidden Markov model

Definition

A hidden Markov model (HMM) is a state-space model whose state is finite.

(Note: this is not a universal definition.)

So let

- $\pi_t^{t'}$ be the probability distribution for the state at time t given information up to time t' , e.g. $\pi_{t,i}^{t'} = P(\theta_t = i | y_{1:t'})$.
- P be the transition probability matrix, e.g. P_{ij} is the probability of moving from state i to state j in 1 time step.
- $p(y_t | \theta_t)$ be the observation density or mass function.

Inference in a hidden Markov model

Assume π_0^0 is given.

- What is forecast distribution at time t given only $\pi_0^0 = \pi_0$, i.e. π_t^0 ?
 Recursively, we have

$$\pi_t^0 = \pi_{t-1}^0 P.$$

Alternatively, we have

$$\pi_t^0 = \pi_0 P^t \quad P^t = P^{t-1} P \quad \text{and} \quad P^1 = P$$

- What is the filtered distribution at time t , i.e. what are the $\pi_{t,s}^t$ for all states s ? Find this recursively via

$$\pi_{t,s}^t \propto p(y_t | \theta_t = s) \pi_{t-1}^{t-1} \cdot P_{\cdot,s}$$

Although smoothing can be useful, it is often of more use in Bayesian analyses to perform backward sampling.

Joint posterior

The joint distribution for $\theta = (\theta_0, \theta_1, \dots, \theta_T)$ can be decomposed as

$$\begin{aligned} p(\theta|y) &= p(\theta_0, \theta_1, \dots, \theta_T|y_{1:T}) \\ &= p(\theta_T|y_{1:T}) \prod_{t=T}^1 p(\theta_{t-1}|\theta_t, y_{1:T}) \\ &= p(\theta_T|y_{1:T}) \prod_{t=T}^1 p(\theta_{t-1}|\theta_t, y_{1:t-1}) \end{aligned}$$

where

$$\begin{aligned} p(\theta_{t-1}|\theta_t, y_{1:t-1}) &= \frac{p(\theta_t|\theta_{t-1}, y_{1:t-1})p(\theta_{t-1}|y_{1:t-1})}{p(\theta_t|y_{1:t-1})} \\ &= \frac{p(\theta_t|\theta_{t-1})p(\theta_{t-1}|y_{1:t-1})}{p(\theta_t|y_{1:t-1})} \\ &\propto p(\theta_t|\theta_{t-1})p(\theta_{t-1}|y_{1:t-1}) \end{aligned}$$

Backward sampling

The joint distribution for θ can be decomposed as

$$p(\theta|y) = p(\theta_T|y_{1:T}) \prod_{t=1}^T p(\theta_{t-1}|\theta_t, y_{1:t-1})$$

and

$$p(\theta_{t-1}|\theta_t, y_{1:t-1}) \propto p(\theta_t|\theta_{t-1})p(\theta_{t-1}|y_{1:t-1}).$$

Suppose we have all the filtered distributions, i.e. π_t^t for $t = 0, \dots, T$.

An algorithm to obtain a joint sample for θ is

1. Sample $\theta_T \sim p(\theta_T|y_{1:T})$ which is a discrete distribution with $P(\theta_T = s|y_{1:T}) = \pi_{T,s}^T$.
2. For $t = T, \dots, 1$, sample θ_{t-1} from a discrete distribution with

$$P(\theta_{t-1} = s|\theta_t, y_{1:t-1}) \propto P_{s,\theta_t} \pi_{T-1,s}^{T-1} = \frac{P_{s,\theta_t} \pi_{T-1,s}^{T-1}}{\sum_{s'=1}^S P_{s',\theta_t} \pi_{T-1,s'}^{T-1}}.$$

Markov model

Consider a Markov model where the states are observed directly, but the transition probability matrix Ψ is unknown. If the sequence of states are $y_{1:t} = (y_1, \dots, y_t)$, we are interested in the posterior

$$p(\Psi|y_{1:t}).$$

Since this is a row stochastic matrix Ψ , we have

$$\sum_{s'=1}^S \Psi_{ss'} = 1 \quad \forall s.$$

So what priors are reasonable for Ψ ?

Priors for row stochastic matrices

One option is a set of independent Dirichlet distributions for each row, i.e. let $\Psi_{s\cdot}$ be the i th row of Ψ , then

$$\Psi_{s\cdot} \sim \text{Dir}(A_s)$$

where A_s is a vector of length S and A is the matrix with rows A_s .

Do we want more structure here?

- sparsity (many zero elements)
- similarity between rows

Dirichlet distribution

The Dirichlet distribution (named after Peter Gustav Lejeune Dirichlet), i.e. $P \sim \text{Dir}(a)$, is a probability distribution for a probability vector of length S . The probability density function for the Dirichlet distribution is

$$p(P; a) = \frac{\Gamma(a_1 + \cdots + a_S)}{\Gamma(a_1) \cdots \Gamma(a_S)} \prod_{s=1}^S p_s^{a_s-1}$$

where $p_s \geq 0$, $\sum_{s=1}^S p_s = 1$, and $a_s > 0$.

Letting $a_0 = \sum_{s=1}^S a_s$, then some moments are

- $E[p_s] = \frac{a_s}{a_0}$,
- $V[p_s] = \frac{a_s(a_0 - a_s)}{a_0^2(a_0 + 1)}$,
- $\text{Cov}(p_s, p_{s'}) = -\frac{a_s a_{s'}}{a_0^2(a_0 + 1)}$, and
- $\text{mode}(p_s) = \frac{a_s - 1}{a_0 - S}$ for $a_s > 1$.

A special case is $S = 2$ which is the beta distribution.

Conjugate prior for multinomial distribution

The Dirichlet distribution is the natural conjugate prior for the multinomial distribution. If

$$Y \sim Mult(n, \pi) \quad \text{and} \quad \pi \sim Dir(a)$$

then

$$\pi|y \sim Dir(a + y).$$

Some possible default priors are

- $a = 1$ which is the uniform density over π ,
- $a = 1/2$ is Jeffreys prior for the multinomial,
- $a = 1/S$ and
- $a = 0$, an improper prior that is uniform on $\log(\pi_h)$. The resulting posterior is proper if $y_h > 0$ for all h .

Dirichlet priors for Markov models

Let A be the hyperparameter with rows A_s such that

$$\Psi_i \stackrel{\text{ind}}{\sim} \text{Dir}(A_s)$$

and C be the count matrix of observed transitions, i.e. C_s is the count vector of transitions from s to all states and $C_{ss'}$ is the count of transitions from s to s' .

The posterior distribution $p(\Psi|y_t)$ is fully conjugate with $A' = A + C$ such that

$$\Psi_s|y \stackrel{\text{ind}}{\sim} \text{Dir}(A'_s) \stackrel{d}{=} \text{Dir}(A_s + C_s)$$

where A'_i is the i th row of A' .

Inference for HMM with unknown transition matrix Ψ

Suppose we have a HMM with unknown transition matrix Ψ . How can we perform posterior inference?

If we assume $\Psi_s \stackrel{\text{ind}}{\sim} \text{Dir}(A)$, then a Gibbs sampling approach is

1. Sample $\theta_{1:t} | \Psi, y \sim \prod_{t=1}^T p(\theta_{t-1} | \theta_t, y_{1:t}, \Psi)$.
2. For $s = 1, \dots, S$, sample $\Psi_s | \theta, y \stackrel{\text{ind}}{\sim} \text{Dir}(A_s + C_s)$ where C_s is the count vector of transitions from s to all states.